

HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		award (1) for either of following - equal number of protons and electrons - both have 12 protons and 12 electrons do not accept any reference to neutrons		1		1		
		(ii)		award (1) for either of following - has more protons than electrons - has 12 protons and 10 electrons neutral answer – it has a 2+ charge		1		1		
		(iii)		they have the same number of protons and different numbers of neutrons neutral answer – X has 2 more neutrons				1		
	(b)	(i)		2,8,1		1		1		
		(ii)		Group 1 Period 3 both needed no ecf possible		1		1		
				Question 4 total	0	5	0	5	0	0